

SKILL WEEK – 12

TASK1: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

```
shreyaj@LAPTOP-FT6VVM4D: ~$ mkdir -p ~/Skill12
cd ~/Skill12
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ ls -l | grep ".txt" | wc -l
ps aux | grep python | wc -l
0
3
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ sleep 10 &
sleep 15 &
sleep 20 &
jobs
ps
[1] 1305
[2] 1311
[3] 1317
[1]-  Running                  sleep 10 &
[2]-  Running                  sleep 15 &
[3]+  Running                  sleep 20 &
PID TTY          TIME CMD
1229 pts/0      00:00:00 bash
1305 pts/0      00:00:00 sleep
1311 pts/0      00:00:00 sleep
1317 pts/0      00:00:00 sleep
1328 pts/0      00:00:00 ps
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ echo "Job 1 started"; sleep 5; echo "Job 1 finished" &
echo "Job 2 started"; sleep 8; echo "Job 2 finished" &
echo "Job 3 started"; sleep 10; echo "Job 3 finished" &
Job 1 started
[1] Done                      sleep 10
[4] 1342
Job 1 finished
Job 2 started
[4]+ Done                      echo "Job 1 finished"
[2]- Done                      sleep 15
[3]+ Done                      sleep 20
[1] 1349
Job 2 finished
Job 3 started
[1]+ Done                      echo "Job 2 finished"
[1] 1358
Job 3 finished
[1]+ Done                      echo "Job 3 finished"
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ top
top - 05:59:07 up 30 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
top - 05:59:09 up 30 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
```

```
[1]+ Done                      echo "Job 2 finished"
[1] 1358
Job 3 finished
[1]+ Done                      echo "Job 3 finished"
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ top
top - 05:59:07 up 30 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
top - 05:59:09 up 30 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7776.6 total, 7118.1 free, 563.7 used, 246.8 buff/cache
top - 05:59:13 up 30 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7776.6 total, 7118.1 free, 563.7 used, 246.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7213.0 avail Mem

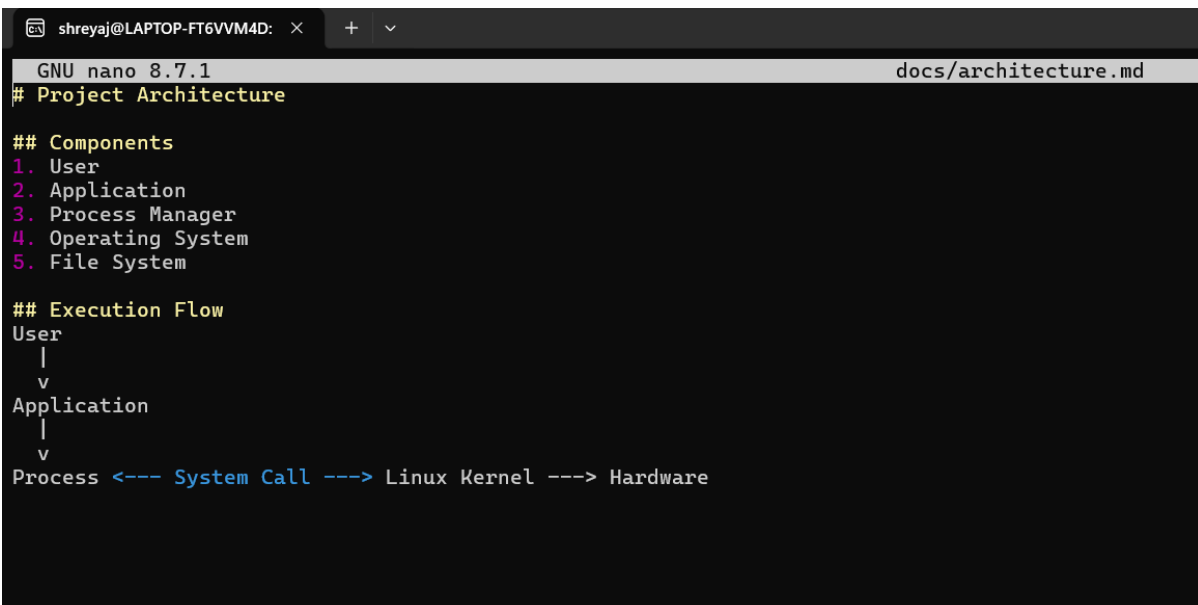
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1370 shreyaj 20   0   8328    6092   3892 R   1.0   0.1   0:00.02 top
    1 root     20   0   24144   15408  11592 S   0.0   0.2   0:01.22 systemd
    2 root     20   0   3180    2204   2072 S   0.0   0.0   0:00.02 init-systemd(Ub
    7 root     20   0   3180    2044   1940 S   0.0   0.0   0:00.00 init
    50 root     19  -1  50404   16876  15572 S   0.0   0.2   0:00.28 systemd-journal
    87 systemd+ 20   0  22416  14444   11956 S   0.0   0.2   0:00.09 systemd-resolve
    94 root     20   0  35436  12356   9192 S   0.0   0.2   0:01.32 systemd-udev
   116 root     20   0   2888    2016   1884 S   0.0   0.0   0:00.05 chronyd-starter
   117 root     20   0   4472    3136   2884 S   0.0   0.0   0:00.01 cron
   118 message+ 20   0   8028    5076   4700 S   0.0   0.1   0:00.12 dbus-daemon
   130 root     20   0  40100   29788  14652 S   0.0   0.4   0:00.33 networkd-dispat
   166 root     20   0  18444    9420   8168 S   0.0   0.1   0:00.10 systemd-logind
   193 syslog   20   0  220548   5860   4576 S   0.0   0.1   0:00.10 rsyslogd
   202 root     20   0   5492    2796   2612 S   0.0   0.0   0:00.01 agetty
   207 root     20   0  123464  32596  17988 S   0.0   0.4   0:00.28 unattended-upgr
   235 _chrony    20   0  23588  11040   7092 S   0.0   0.1   0:00.10 chronyd
   239 _chrony    20   0  12096    2408   1760 S   0.0   0.0   0:00.00 chronyd
   340 root     20   0   8648    5560   4716 S   0.0   0.1   0:00.04 login
   388 shreyaj 20   0  22760  13368  10920 S   0.0   0.2   0:00.28 systemd
   390 shreyaj 20   0  22916    4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
   418 shreyaj 20   0   6116    5508   3912 S   0.0   0.1   0:00.06 bash
   896 polkitd   20   0  306408   8080   7668 S   0.0   0.1   0:00.13 polkitd
  1222 root     20   0   3184    1116    980 S   0.0   0.0   0:00.00 SessionLeader
  1223 root     20   0   3200    1256   1108 S   0.0   0.0   0:00.00 Relay(1229)
  1229 shreyaj 20   0   6784    6148   3936 S   0.0   0.1   0:00.06 bash
```

TASK2: To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ mkdir -p docs/design docs/diagrams
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano docs/architecture.md
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano docs/diagrams/flow.txt
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano docs/execution_path.md
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano docs/design/design_decisions.md
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ find docs -type f
cat docs/architecture.md
docs/design/design_decisions.md
docs/architecture.md
docs/diagrams/flow.txt
docs/execution_path.md
# Project Architecture

## Components
1. User
2. Application
3. Process Manager
4. Operating System
5. File System

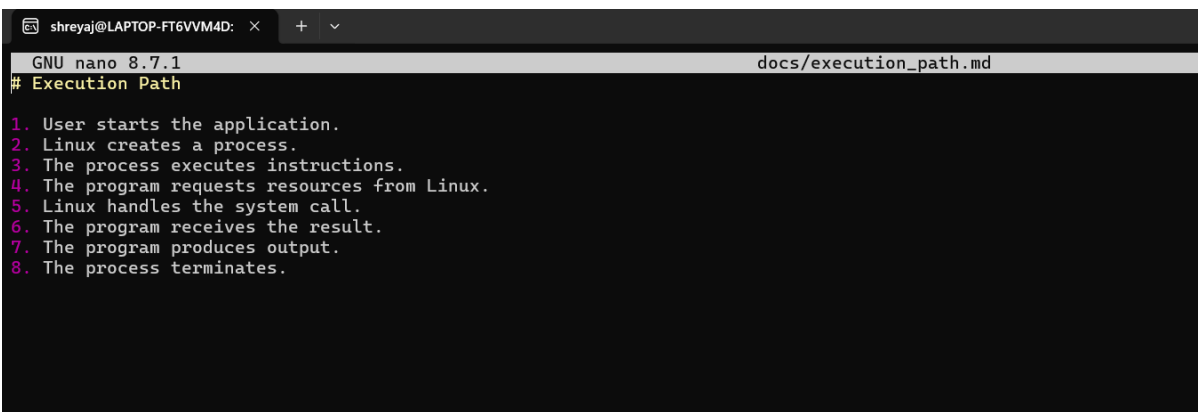
## Execution Flow
User
|
v
Application
|
v
Process <--- System Call ---> Linux Kernel ---> Hardware
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$
```



```
shreyaj@LAPTOP-FT6VVM4D: × + v
GNU nano 8.7.1 docs/architecture.md
# Project Architecture

## Components
1. User
2. Application
3. Process Manager
4. Operating System
5. File System

## Execution Flow
User
|
v
Application
|
v
Process <--- System Call ---> Linux Kernel ---> Hardware
```



```
shreyaj@LAPTOP-FT6VVM4D: × + v
GNU nano 8.7.1 docs/execution_path.md
# Execution Path

1. User starts the application.
2. Linux creates a process.
3. The process executes instructions.
4. The program requests resources from Linux.
5. Linux handles the system call.
6. The program receives the result.
7. The program produces output.
8. The process terminates.
```

```
shreyaj@LAPTOP-FT6VVM4D: × + v
GNU nano 8.7.1 docs/diagrams/flow.txt
Start
|
v
Read Data
|
v
Process Data
|
v
System Call ---> Linux Kernel ---> Hardware
|
v
Check Errors
|
v
Output
|
v
End
```

```
shreyaj@LAPTOP-FT6VVM4D: × + v
GNU nano 8.7.1 docs/design/design_decisions.md
# Design Decisions

## Decision 1
We use Linux processes to execute independent jobs.

## Decision 2
Pipelines are used to transfer output between commands.

## Decision 3
Logs are stored for detecting failures.

## Decision 4
System resources are monitored using Linux utilities.

## Decision 5
Git is used for source code management.
```

TASK3: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

```
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ mkdir -p src tests scripts
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano src/main.c
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano build.sh
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ chmod +x build.sh
./build.sh
./build/main
Building project...
Build successful!
Project executed successfully!
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ nano README.md
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ ./build/main
Project executed successfully!
```

```
shreyaj@LAPTOP-FT6VVM4D: x + v - □ ×
GNU nano 8.7.1 src/main.c
#include <stdio.h>

int main() {
    printf("Project executed successfully!\n");
    return 0;
}
```

```
shreyaj@LAPTOP-FT6VVM4D: x + v - □ ×
GNU nano 8.7.1 build.sh
#!/bin/bash

echo "Building project..."
mkdir -p build
gcc src/main.c -o build/main

if [ $? -eq 0 ]; then
    echo "Build successful!"
else
    echo "Build failed!"
    exit 1
fi
```

TASK4: To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

```
shreyaj@LAPTOP-FT6VVM4D: x + v - □ ×
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ ls -l | grep ".txt" | wc -l
0
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ sleep 100 &
jobs
[1] 1704
[1]+  Running                  sleep 100 &
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ kill -STOP 1388
jobs
kill -CONT 1388
jobs
kill -9 1388
-bash: kill: (1388) - No such process
[1]+  Running                  sleep 100 &
-bash: kill: (1388) - No such process
[1]+  Running                  sleep 100 &
-bash: kill: (1388) - No such process
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ ./build.sh
./build/main
Building project...
Build successful!
Project executed successfully!
shreyaj@LAPTOP-FT6VVM4D:~/Skill12$ cat README.md
git log --oneline
# Linux Project

## Description
This project demonstrates Linux processes, pipelines, system calls, monitoring, and performance analysis.

## Directory Structure
- `src`: Source code
- `tests`: Test files
- `scripts`: Utility scripts
- `docs`: Documentation
- `build`: Compiled files

## Build
Run:
``bash
./build.sh
fatal: not a git repository (or any of the parent directories): .git
```